

The Game Master's Binder Reference Booklet

A Practical Blueprint and Structural Guide to Upgrading Legacy Maps to Dynamic VTT Specifications

Welcome to your physical reference booklet for the **UVTT v2 Upgrader Web SPA**. This booklet translates the system's high-performance, CAD-style workspace into practical blueprints. It is designed to sit in your GM binder so you can quickly configure walls, portals, lighting, spatial soundscapes, and automation links during offline map design.

1. CORE RUNTIMES: WEB VS. DESKTOP PRO

While both environments share the exact same UI and drafting tools, they handle your computer's hardware resources very differently. Choose the appropriate runtime based on your campaign's map scale and resource complexity:

Feature Parameter	Web Application (Browser SPA)	Desktop Pro (Native Client)
File Ingestion Method	Manual drag-and-drop or file upload into browser window due to strict HTML5 security protocols.	Natively mounts your local hard drive OS folder; select your master asset folder once to link everything.
Streaming & RAM Scale	Constrained by strict browser-enforced memory caps. Best suited for quick edits or single-layer maps.	Natively streams gigabytes of high-resolution textures, tokens, and audio directly from your local drive.
Optimal Use-Case	Single-floor dungeons, quick touch-ups, or editing on-the-go on any machine.	Massive multi-story megadungeons, city maps, and heavy spatial audio soundscapes. Prevents RAM crashes.

2. SCENIC TOUR OF THE WORKSPACE

The interface is mapped into five dedicated zones to keep the drafting layout completely clean:

- The Canvas Viewport (Center):** High-performance, WebGPU-accelerated rendering canvas. Floating in the bottom right corner of this canvas is the *Toggle Grid Coordinate Cursor* button, which tracks your exact mathematical X/Y coordinates as you move your mouse.
- The Top Bar (Active Floor & History):** Anchored at the top center. Houses the floor switcher (globe icon & dropdown), map layer renamer, Undo/Redo buttons, and controls to add blank layers, import legacy maps (.dd2vtt/.df2vtt), or delete active floors.
- The Global Panel (Top Right):** The master control command center. Houses collapsible menus for *File & Export*, *Map Levels* (Catalog), *Format & Platform Exports*, *Action History*, *Environment Config*, and *Global Audio*.
- The Export Manager (Bottom Right):** Houses the *Export Profile* dropdown and the prominent green **Run Compile & EXPORT** button to assemble your final game package.
- The Left Drafting Toolbar:** Houses specialized sub-panels containing your drafting toolkit. *Architecture*: structural layout tools. *Entities*: dynamic, interactive layers. *Assets (Desktop Pro Only)*: native asset library. *Environment*: visualization toggles.

II. THE STRUCTURAL TOOLKIT (ARCHITECTURAL DRAFTING)

Architectural drafting tools define the boundaries of exploration. Line geometries, collision zones, and overhead canopies are constructed using coordinates and mathematical snapping rules.

1. THE WALL TOOL (PHYSICAL GEOMETRY & VISION)

Walls block or guide lines of sight and token movement. Selecting the tool with no wall selected configures placing defaults, while clicking an existing wall activates Editing Mode, revealing clipboard utilities (Copy, Paste, Duplicate, Delete) and Player View visibility overrides.

- **Apply Smoothing Pass:** Mathematically curves linear walls. *Note: This utility only enables if the selected wall contains at least three vertices (two segments). Single straight lines cannot be curved.*
- **Collision Presets:** Instantly define how tokens and light interact. Use **Solid** for load-bearing dungeon walls, or **Ethereal** for low fences, illusory walls, or lines that block sight but allow movement.
- **Directional Blocking (One-Way Walls):** Define vision blocking that works from one side only. This is configured using the *Right-Hand Rule*: vision is blocked looking from the left of the line's drawing direction to the right, or vice versa, based on the drawn vertices.
- **Z-Height Bounds:** Define the exact vertical floor (bottom) and ceiling (top) elevations of the wall in grid units to support verticality.
- **Snap All Vertices:** Instantly aligns wall coordinates to the active grid. Can be run globally or on selected walls in Editing Mode.

2. THE PORTAL TOOL (DOORS, WINDOWS, & PASSAGES)

Portals define structural elements that can change state during gameplay (opening, closing, locking). You can bulk-edit selected portals to swap types or properties instantly.

- **Portal Architecture:** Dropdown to define the physical structure: **Door** (standard player toggle), **Window** (blocks movement but permits light and vision), **Secret Door** (invisible to players on VTT until discovered), or **Portcullis**.
- **Initial State:** Sets how the entryway spawns in the virtual tabletop when loaded: **Open**, **Closed**, or **Locked**.
- **Z-Height Bounds (Elevation Ranges):** Perfect for placing high dungeon slit windows, arrow slits, or elevated cage grates that only block sight and movement at specific elevations.

3. THE ROOF TOOL (OVERHEAD CANOPIES & MULTI-STORY LAYERS)

Roofs define overhead bounding polygons (such as forest canopies, ceiling covers, or second-story floors). In compliant Virtual Tabletops, these zones dynamically fade or completely hide the moment a player's token steps underneath them.

- **Roof Tint/Color & Opacity (%):** Define how the roof renders for the GM (or on the map before tokens step inside). Adjust opacity to make tree canopies translucent or roofs fully opaque.
- **Z-Height Bounds:** Defines the top and bottom limits of the overhead canopy (essential for multi-tier towers where players can stand on top of a roof or run beneath it).
- **Hidden from Players:** Keeps the overhead zone invisible to the players while keeping it fully visible and manageable for the GM.

4. GRID ALIGNMENT: THE 'RUBBER SHEETER'

Aligning baked-in map art grids with VTT coordinate systems is known as *Rubber Sheeting*. Selecting the Grid Alignment tool opens a calibration panel providing two distinct alignment workflows:

The Automated 3-Step Workflow (Highly Recommended):

- **1. Pin Origin:** Click the top-left intersection of any grid square on the raw map artwork to lock the mathematical origin point.
- **2. Drag Auto-Align Box:** Click and drag a green bounding box over a single 1x1 map square on the artwork.
- **3. Calculate:** The Upgrader automatically calculates the optimal pixel-per-grid-unit DPI and offset coordinates.

Manual Fine-Tuning Calibration:

- **Grid X/Y (DPI) & X/Y Offset (px):** Input boxes to manually scale (DPI) or shift the grid starting point along horizontal or vertical axes.
- **Micro-Nudge Buttons:** Click → **Step +X**, ← **Step -X**, ↓ **Step +Y**, or ↑ **Step -Y** to nudge the grid by single-pixel increments.

III. THE INTERACTIVE TOOLKIT (DYNAMIC ENTITIES & TRAPS)

Entities are placing anchors that bring the environment to life with light, sound, spawn points, and mechanical automation. When you select any active entity on the canvas, the system prepends universal CAD controls: **Clipboard** (Copy, Paste, Duplicate, Delete), **Visibility Overrides** (force show/hide in Player View), and **Quick Actions** (Clone/Delete).

1. DYNAMIC LIGHTS & SPATIAL AUDIO

Configure visual ambiance and acoustic environments using precise spatial falloffs:

The Light Tool:

- **Lighting Projection Type:** Choose between **Point** (360-degree omnidirectional glow) and **Cone/Beam** (directional headlights).
- **Z-Axis Elevation & Radius:** Define floating height in grid units, and configure **Bright Radius** vs. **Dim Radius** boundaries.
- **Hex Color, Intensity & Decay:** Select exact hex colors. Determine light fade styles using the **Decay Model** (Inverse Square vs. Linear).
- **Animation Profile:** Inject environmental movement with animated presets: *flicker* (torches), *pulse* (runes), or *strobe* (alarms).

The Audio Tool:

- **Acoustic Falloff:** Set **Inner Core** (radius of constant 100% volume) and **Max Range** (outer limit where sound fades linearly to 0%).
- **Muffled by Walls (Occlusion):** A critical checkbox. Tells the VTT engine to automatically muffle/attenuate the spatial audio if a drawn wall breaks the direct line of sight between the player token and the sound source.

2. SPAWN POINTS, EMITTERS, & GLOBAL SCHEMATICS

Manage player entryways and dynamic climate effects:

The Spawn Tool:

- **Token Arrival Heading:** A slider (0 to 360 degrees) to set the exact direction tokens will face upon spawning.
- **Default Landing Zone:** *Mandatory Checkbox*. Every map layer must have exactly one designated Default Landing Zone so the VTT knows where to place newly entering player characters.

The Emitter Tool (Particle Weather):

- **Map-Wide vs. Local:** Weather can fill the entire canvas (blizzards) or be contained within defined rooms (basement fog).
- **Weather Styles:** Particle presets including **Rain**, **Snow**, **Embers**, and **Fog**.
- **Z-Index Layering & Particle Sliders:** Set particle rendering height (above or below roofs) and use sliders for *Scale (%)*, *Direction*, *Speed*, *Intensity (Density)*, and *Variance (Fluctuation %)*.

3. THE EVENT TOOL & WIRE TARGETS (THE TRAPMASTER'S BLUEPRINT)

The Event Tool builds trigger-response interactions without coding. Triggers are physically linked to targets (like locked doors, portals, or lights) using the **Wire Targets** system:

- **Activation Methods:** Set how the trigger fires: **On Step** (token enters zone), **On Click** (player interacts), or **GM Only** (invisible trigger operated solely from the GM overlay).
- **Target Actions:** Determine what happens when fired: **Teleport**, **Toggle Visibility**, or **Play Sound**.
- **Establishing the Wire Link:** Select *both* the Event Trigger and the Target Object (e.g., a Door) on the canvas. The **Wire Targets** panel will slide open in the floating panel. Click **Bind Selected Entities** to establish the wire link. To dismantle the connection, click **Clear Targets**.

WIRE TARGETS FUNCTIONAL FLOW SCHEMATIC:

```
[ PLAYER TOKEN ] --(steps on)--> [ EVENT TRIGGER (On Step) ]
|
[ BIND WIRE LINK ]
|
v
```

[TRIGGER ACTION] -----> [PORTAL TARGET: OPEN DOOR]

IV. GLOBAL LAYERS, CACHE MANAGEMENT, & EXPORT WORKFLOWS

1. MAP CATALOG & FOUNDATIONAL GRID VARIABLES

Configure underlying grid mathematics and layer structures:

- **Grid Topology:** Select grid layouts: standard **Square**, hexagonal grids (**Hex Horizontal** / **Hex Vertical**), or **Isometric** layouts.
- **Grid Size & Color:** Explicitly define pixel-per-grid resolutions (e.g., 70px or 100px) and customize visual grid colors.
- **Main & Subgrid Thickness Sliders:** Separately adjust visual weights of macro grids (5-foot movement squares) and micro grids (used for high-precision vertex snapping).
- **Map Levels (The Catalog):** View multi-floor layers in a structured list. Rename layers with the **Pencil (Edit)** icon, or permanently purge structures using the **Trash Can (Delete)** button.

■■ CRITICAL RAM SAVIOR (Action History Cache Control):

The Upgrader saves a complete snapshot of your map's data with *every single edit* to feed its robust Undo/Redo state machine. During long, detailed drafting sessions, this state history will consume massive amounts of your web browser's allocated RAM, leading to out-of-memory crashes.

GM Action: Open the Global Panel's **Action History** and click the **Clear History** button. This flushes the undo cache, instantly freeing browser memory without affecting your current active map layout. Flush the cache every 45-60 minutes during heavy structural editing.

2. THE EXPORT MANAGEMENT WORKFLOW

Compiling translates your working database into VTT-readable files. Use these options in the Global Panel and Export Manager to package your project:

- **Save & Load Project:** Always save your raw project file. This file preserves raw Bezier curve vectors, original textures, and event wiring so you can re-load and edit your layout in future sessions.
- **Package Catalog as Compound Dungeon:** *Checkbox*. When checked, the compiler bundles all floors in your catalog into a single **.uvtt2z ZIP archive** and automatically rewrites all teleporter links to route internally. When unchecked, it exports levels in **Federated Mode** as independent files.
- **Format & Platform Compile Rules:**
 - *Export v1 (Downgrade):* Converts modern v2 maps back into legacy **.dd2vtt** structures.
 - *Compile v2 (Standard):* Standard compilation to the Universal Virtual Tabletop v2 specification.
 - *Secure Archive (.zip):* Compiles your project into a secured **.uvtt2z ZIP**, decoupling structural data from premium assets and generating cryptographically secure **manifest.hash** files to protect intellectual property.
 - *Platform Exports:* Tailors output format for **Foundry VTT**, **Roll20**, or **Fantasy Grounds**.
- **The Export Manager (Bottom Right):** Set your target profile using the **Export Profile Dropdown**, then hit the prominent green **Run Compile & EXPORT** button to download your playable map file.